

Modeling the Lighting in Scenes as Style for Auto White-Balance Correction

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Motivation

- ◆ Style *may* refer to different concepts depending on how the feature space is formed. Examples:
 - ◆ The haircut style in a face image
 - ◆ Painting style of an artist
 - ◆ Effects applied by social media filters
- ◆ Based upon the findings in previous studies, one may argue that **any disruptive or modifying factors** for the whole image can be modeled as **the style factor**.
- ◆ We can assume that the illumination in the scene basically stands for *the additional style information* injected into the scene.

Learning Weighting Maps by Style

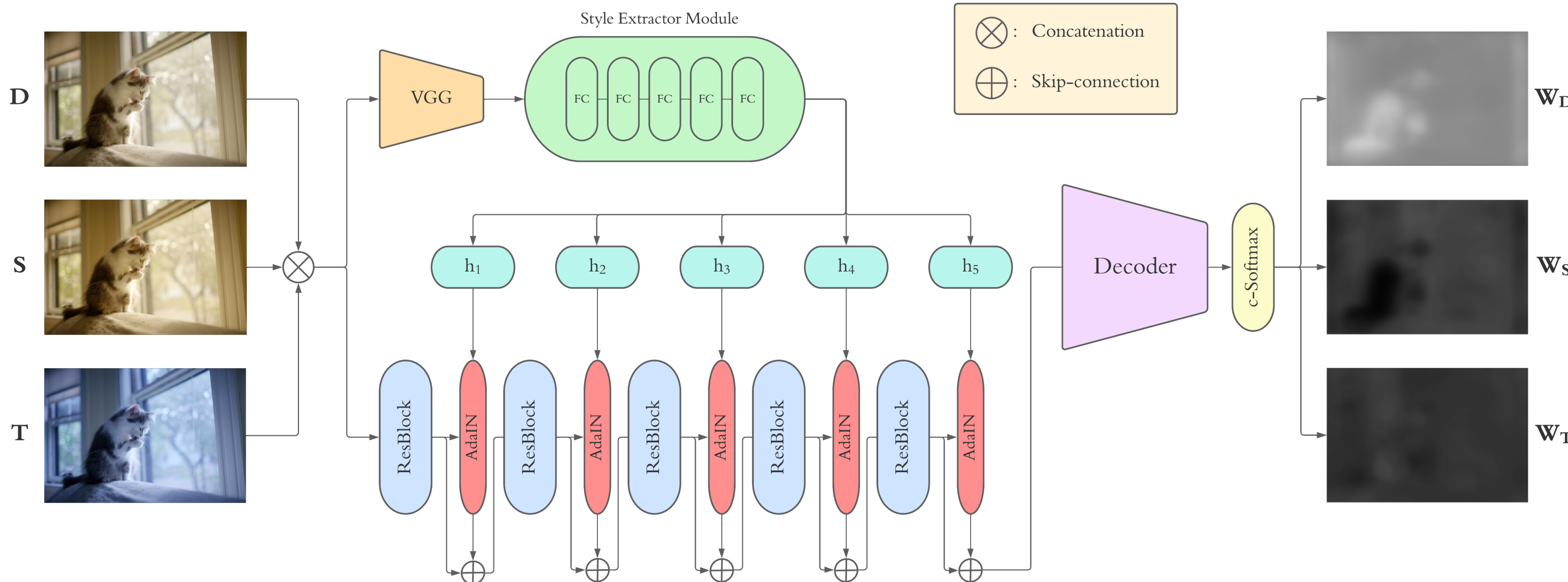


Figure 1: Overall design of proposed learning mechanism for the weighting maps. The latent representations of the images with different WB settings are fed into the style extractor module, then the affine parameters are computed, and sent to the corresponding adaptive feature normalization layer to discard the external style from the feature maps. The decoder part generates the weighting maps for all WB settings. Finally, we blend the final sRGB output by the scenes with different WB settings and their corresponding weighting maps.

Modified Camera ISP

$$\hat{I}_{c_i} = M_{c_i} \phi(I_{init})$$

I_{init} is the initial HR image rendered with a fixed WB setting, $\hat{I}_{c_i}$ represents the output image mapped to the target WB setting. M_{c_i} is the matrix mapping the colors of the initial image by a polynomial kernel function.

$$\tilde{I}_{corr} = \sum_i W_i \odot \tilde{I}_{c_i}$$

I_{corr} is the corrected LR sRGB image, W_i represents the weighting map for i^{th} WB setting (*i.e.*, c_i), and $\sim I$ denotes the small image rendered with c_i .

Single Illuminant Scenarios

Method	MSE				MAE				ΔE 2000				Size
	Mean	Q1	Q2	Q3	Mean	Q1	Q2	Q3	Mean	Q1	Q2	Q3	
FC4 [45]	371.90	79.15	213.41	467.33	6.49°	3.34°	5.59°	8.59°	10.38	6.60	9.76	13.26	5.89 MB
Quasi-U CC [13]	292.18	15.57	55.41	261.58	6.12°	1.95°	3.88°	8.83°	7.25	2.89	5.21	10.37	622 MB
KNN WB [5]	194.98	27.43	57.08	118.21	4.12°	1.96°	3.17°	5.04°	5.68	3.22	4.61	6.70	21.8 MB
Interactive WB [3]	159.88	21.94	54.76	125.02	4.64°	2.12°	3.64°	5.98°	6.20	3.28	5.17	7.45	38 KB
Deep WB [2]	80.46	15.43	33.88	74.42	3.45°	1.87°	2.82°	4.26°	4.59	2.68	3.81	5.53	16.7 MB
Mixed WB [4] results													
$p = 64$, WB={t, d, s}	168.38	8.97	19.87	105.22	4.20°	1.39°	2.18°	5.54°	5.03	2.07	3.12	7.19	5.09 MB
$p = 64$, WB={t, f, d, c, s}	161.80	9.01	19.33	90.81	4.05°	1.40°	2.12°	4.88°	4.89	2.16	3.10	6.78	5.10 MB
$p = 128$, WB={t, f, d, c, s}	176.38	16.96	35.91	115.50	4.71°	2.10°	3.09°	5.92°	5.77	3.01	4.27	7.71	5.10 MB
Our results													
$p = 64$, WB={t, d, s}	92.65	6.52	14.23	35.01	2.47°	0.82°	1.44°	2.49°	2.99	1.36	2.04	3.32	61.0 MB
$p = 64$, WB={t, f, d, c, s}	151.38	29.49	56.35	125.33	4.18°	2.13°	3.03°	4.81°	5.42	3.11	4.42	6.76	61.1 MB
$p = 128$, WB={t, d, s}	88.03	7.92	17.73	45.01	2.61°	0.93°	1.58°	2.85°	3.24	1.50	2.30	3.95	61.2 MB
$p = 128$, WB={t, f, d, c, s}	100.24	10.77	37.74	70.18	3.09°	1.15°	2.61°	3.87°	3.96	1.59	3.55	5.51	61.3 MB

Table 1: Benchmark on single-illuminant Cube+ dataset. Different WB settings are denoted as follows. *t*: tungsten, *f*: fluorescent, *d*: daylight, *c*: cloudy, *s*: shade. *p* refers to the patch size.

Highlights: **the best**, **the second**, **the third**.

Multiple Illuminant Scenarios

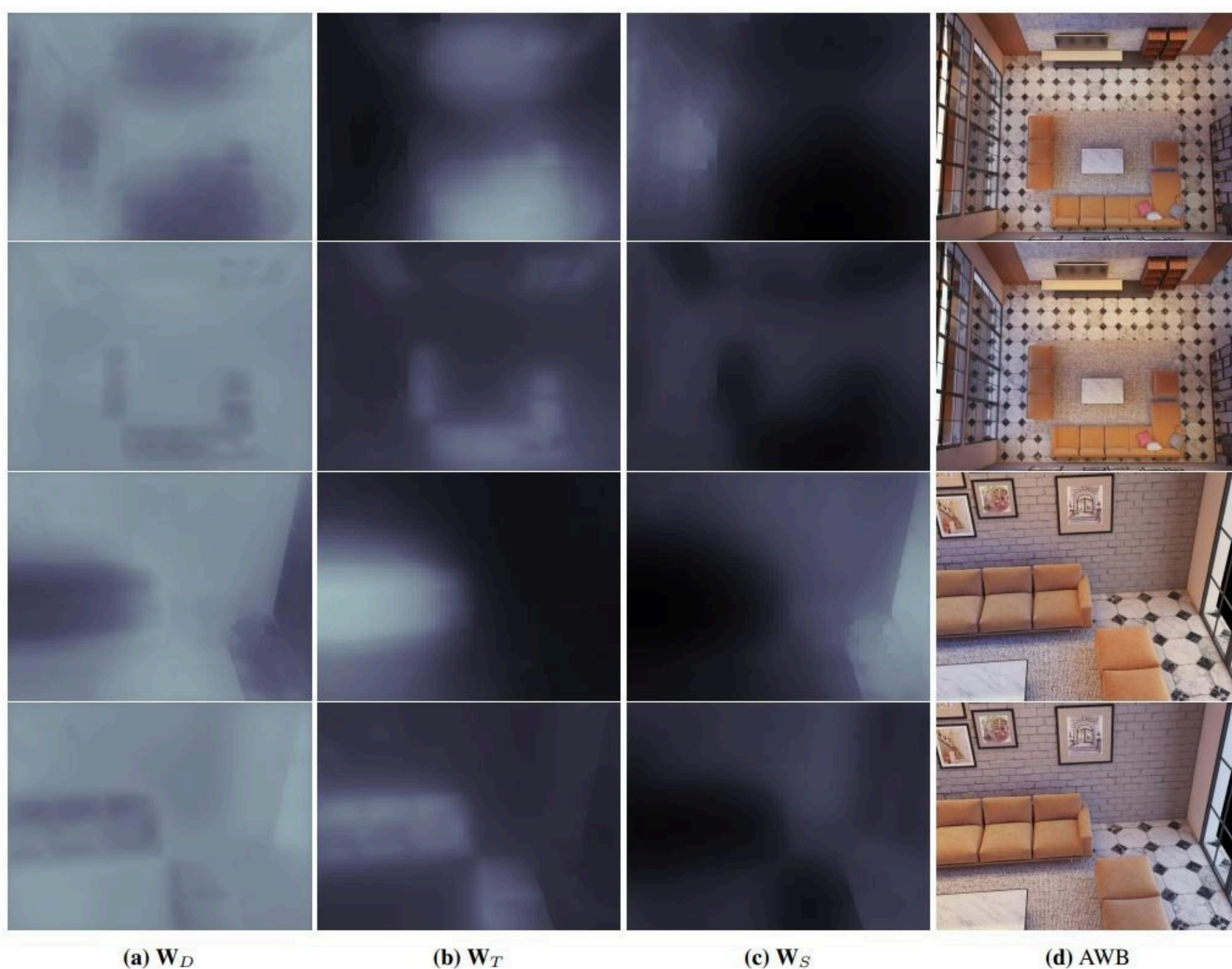


Figure 4: Qualitative comparison of the performance of the prior work (*odd rows*) and our method (*even rows*) on mixed-illuminant dataset.

Ablations

Models	MSE	MAE	ΔE 2000
Single-illuminant dataset, WB = {t, d, s}, $p = 64$			
$ms = 0, eas = 0$	98.55	2.71°	3.32
$ms = 1, eas = 0$	93.78	2.59°	3.15
$ms = 0, eas = 1$	97.20	2.66°	3.28
$ms = 1, eas = 1$	92.65	2.47°	2.99
Mixed-illuminant dataset, WB = {t, d, s}, $p = 128$			
$ms = 0, eas = 0$	878.58	5.05°	12.12
$ms = 1, eas = 0$	843.50	5.04°	11.70
$ms = 0, eas = 1$	843.64	5.04°	11.98
$ms = 1, eas = 1$	822.77	5.11°	11.65

Table 3: The ablation study on using multi-scale weighting maps (*ms*) and applying edge-aware smoothing to weighting maps (*eas*).

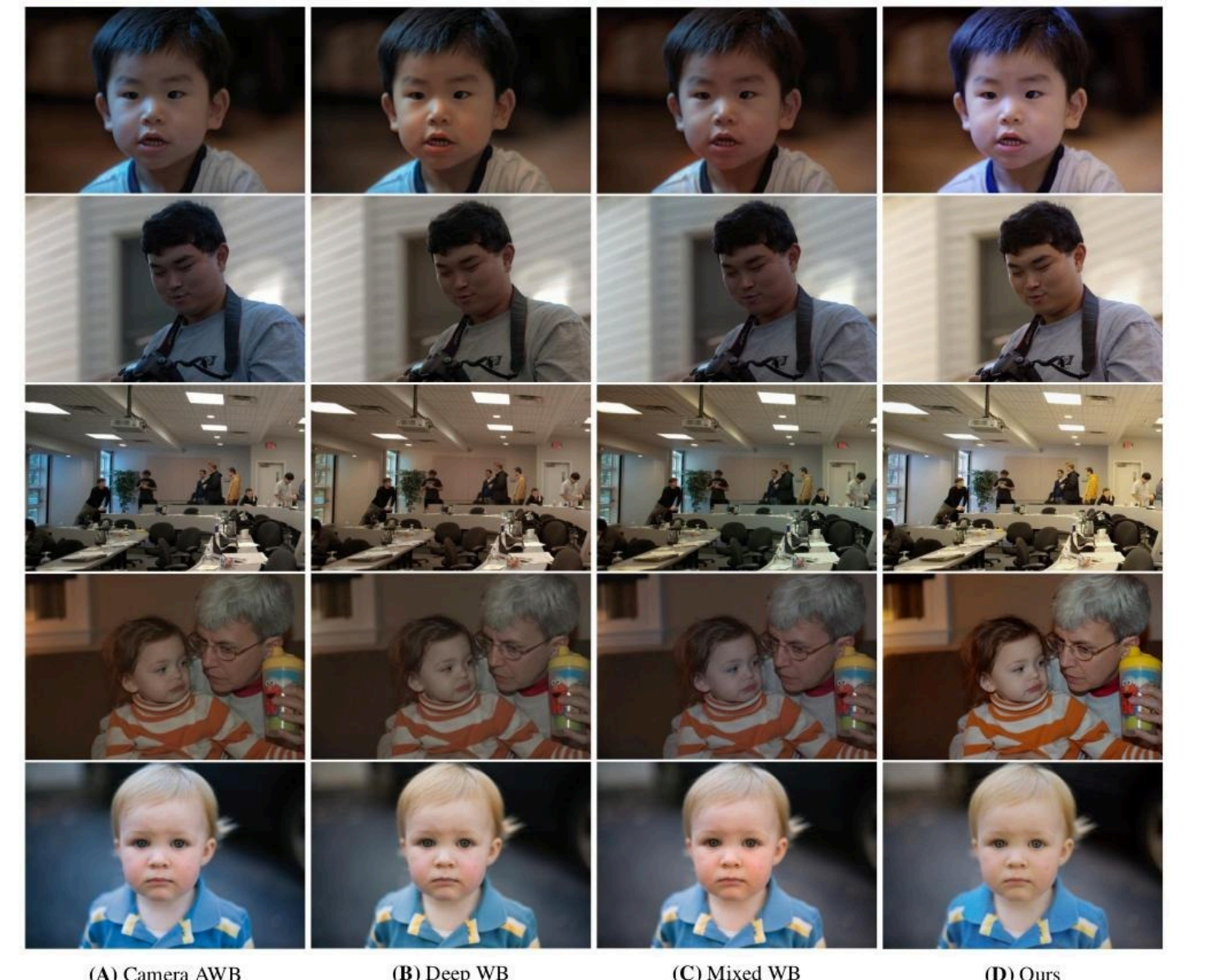


Figure 3: Visual comparison of the results of our method and the others on the selected samples from MIT-Adobe FiveK dataset. Indices from top to bottom: 2882, 606, 659, 2431, 2550.

Method	MSE				MAE				ΔE 2000			
	Mean	Q1	Q2	Q3	Mean	Q1	Q2	Q3	Mean	Q1	Q2	Q3
Gray Pixel [60]	4959.20	3252.14	4209.12	5858.69	19.67°	11.92°	17.21°	27.05°	25.13	19.07	22.62	27.46
Grayness index [59]	1345.47	727.90	1055.83	1494.81	6.39°	4.72°	5.65°	7.06°	12.84	9.57	12.49	14.60
KNN WB [5]	1226.57	680.65	1062.64	1573.89	5.81°	4.29°	5.76°	6.85°	12.00	9.37	11.56	13.61
Interactive WB [3]	1059.88	616.24	896.90	1265.62	5.86°	4.56°	5.62°	6.62°	11.41	8.92	10.99	12.84
Deep WB [2]	1130.60	621.00	886.32	1274.72	4.53°	3.55°	4.19°	5.21°	10.93	8.59	9.82	11.96
Mixed WB [4] results												
$p = 64$, WB={t, d, s}	819.47	655.88	845.79	1000.82	5.43°	4.27°	4.89°	6.23°	10.61	9.42	10.72	11.81
$p = 64$, WB={t, f, d, c, s}	938.02	757.49	961.55	1161.52	4.67°	3.71°	4.14°	5.35°	12.26	10.80	11.58	12.76
$p = 128$, WB={t, d, s}	830.20	584.77	853.01	992.56	5.03°	3.93°	4.78°	5.90°	11.41	9.76	11.39	12.53
$p = 128$, WB={t, f, d, c, s}	1089.69	846.21	1125.59	1279.39	5.64°	4.15°	5.09°	6.50°	13.75	11.45	12.58	15.59
Our results												
$p = 64$, WB={t, d, s}	868.01	649.36	889.00	1026.98	5.73°	4.48°	5.42°	6.34°	12.11	10.42	12.12	13.36
$p = 64$, WB={t, f, d, c, s}	1051.07	760.86	1024.00	1332.50	6.30°	4.43°	6.01°	7.69°	14.43	11.90	13.11	16.15
$p = 128$, WB={t, d, s}	822.77	576.52	840.67	1025.26	5.11°	3.93°	4.85°	5.51°	11.65	10.63	11.86	13.02
$p = 128$, WB={t, f, d, c, s}	834.28	629.95	842.71	1005.59	5.71°	4.57°	5.54°	6.19°	11.79	9.84	12.19	13.00

Table 2: Benchmark on mixed-illuminant evaluation set. Highlights and symbols are the same as in Table 1.

Night Photography Rendering



Figure 5: Visual comparison on the night photography rendering results of the standard camera pipeline, the prior work and our method.